

Rewilding and Detecting Change

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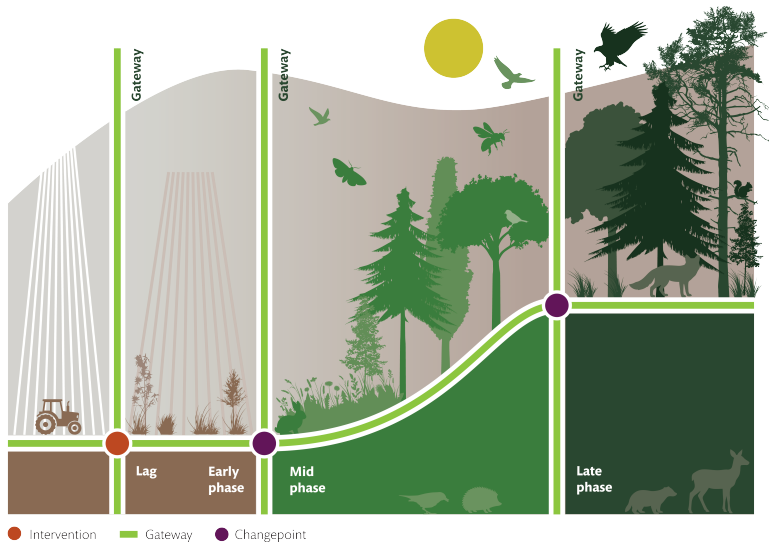
Joint work with

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J. Bull, L. Borger (Swansea)

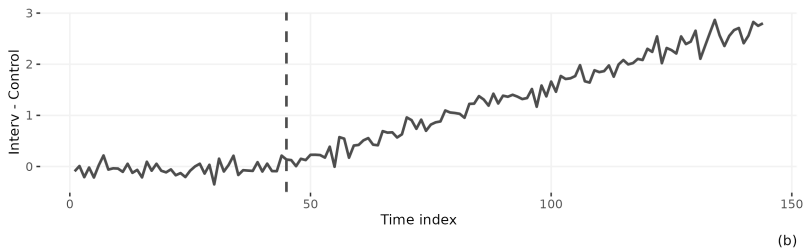
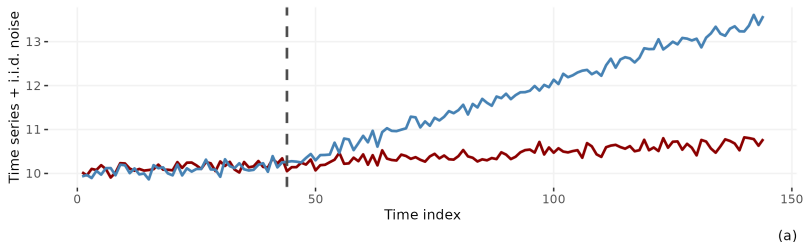
Forestry & Ecology Seminar Sep 2026

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- Rewilding
- What are changepoints?
- Why are they useful?
- Examples



Experiment Setup



Before After Control Intervention

What are changepoints?

Changepoints are also known as:

- breakpoints
- segmentation
- structural breaks
- regime switching
- detecting disorder

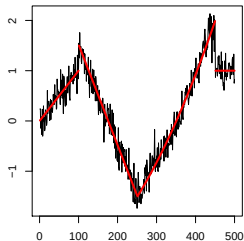
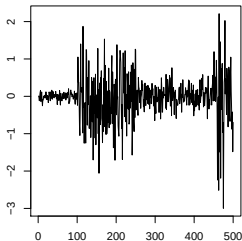
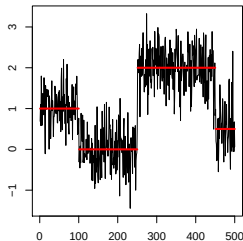
and can be found in a wide range of literature including

- quality control
- finance
- medicine
- environment
- linguistics
- ...

What are changepoints?

For data $y_1, \dots, y_n$, if a changepoint exists at τ , then $y_1, \dots, y_\tau$ differ from $y_{\tau+1}, \dots, y_n$ in some way.

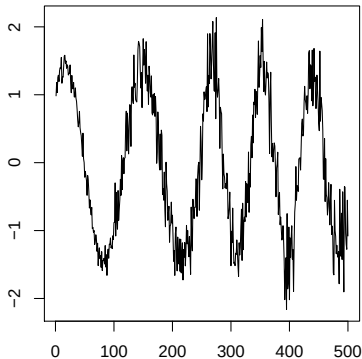
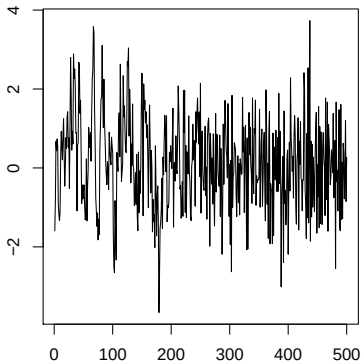
There are many different types of change.



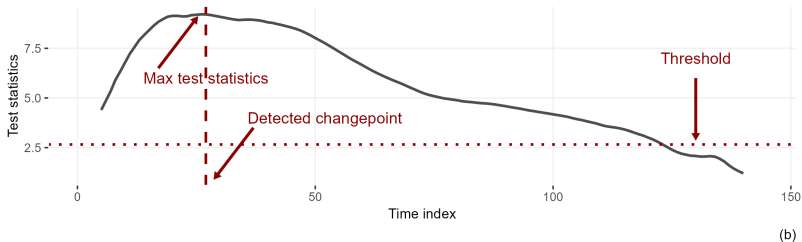
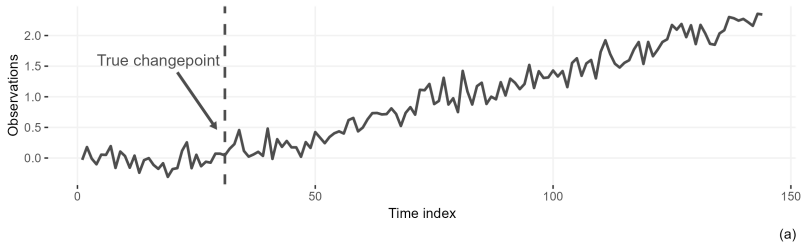
What is the goal?

- Has a change occurred?
- If yes, where is the change?
- What is the difference between the pre and post change data?
 - Maybe this is the type of change
 - Maybe it is the parameter values before and after the change
- What is the probability that a change has occurred?
- How certain are we of the changepoint location?
- How many changes have occurred (+ all the above for each change)?
- Why has there been a change?

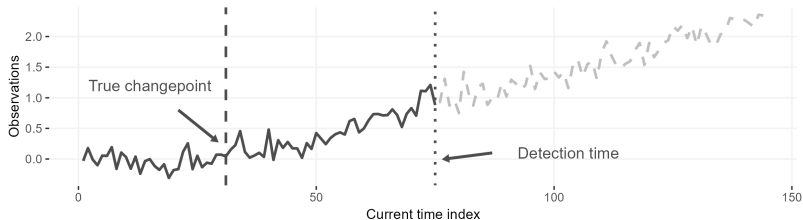
More complicated types of change



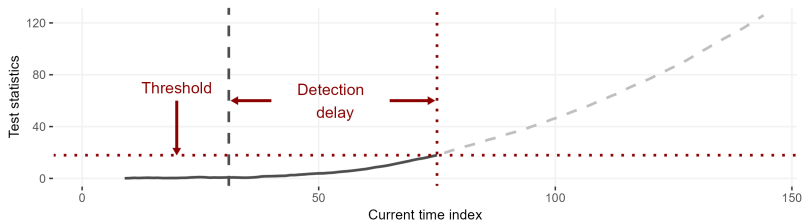
Offline Changepoint Models



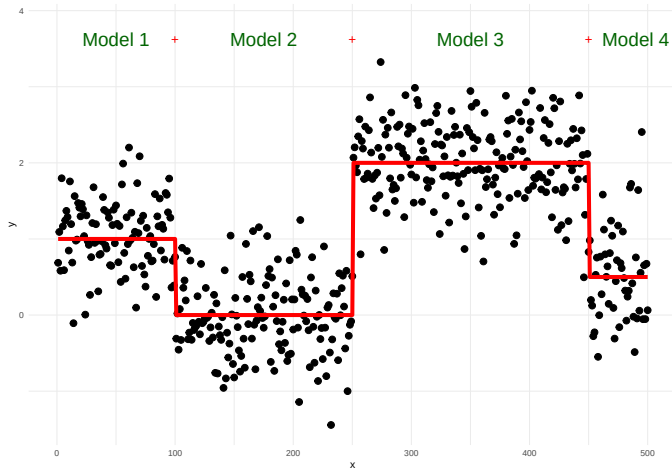
Online Changepoint Models



(a)



(b)

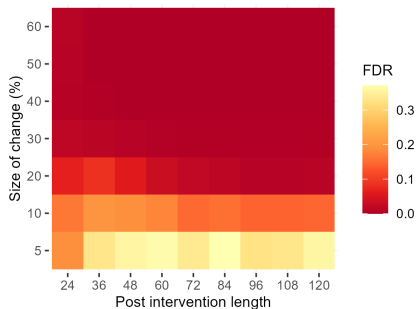
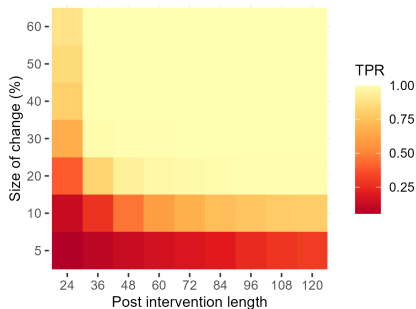


- How many changes?
- Where are the changes? (2^{n-1} possible solutions!)

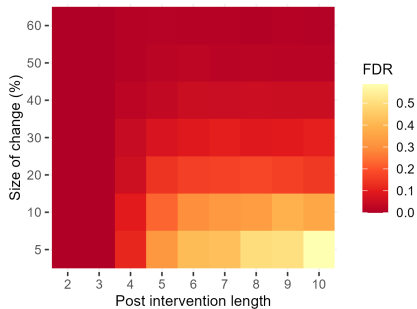
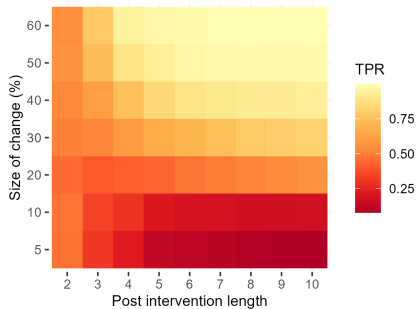
- Optimization
- Model development
- Theoretical properties
- Limits of detection
- Small sample and large sample in the same data

- Assumptions
- Computational Complexity
- Determining type of change
- Quickest / Accurate detection
- Outliers

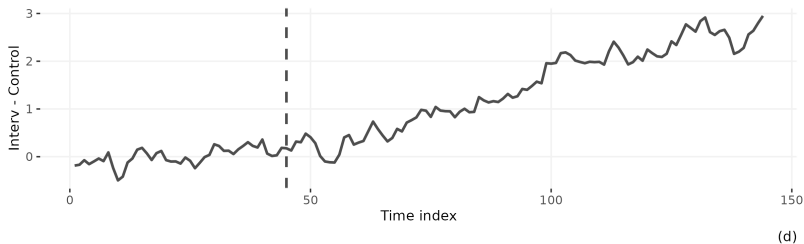
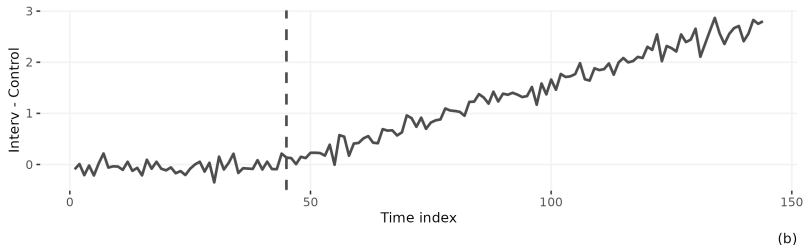
Value of Patience



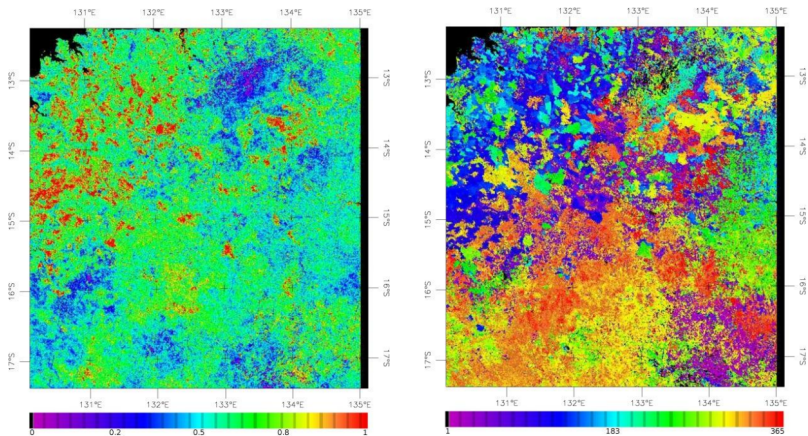
Value of Patience



Effect of Dependence

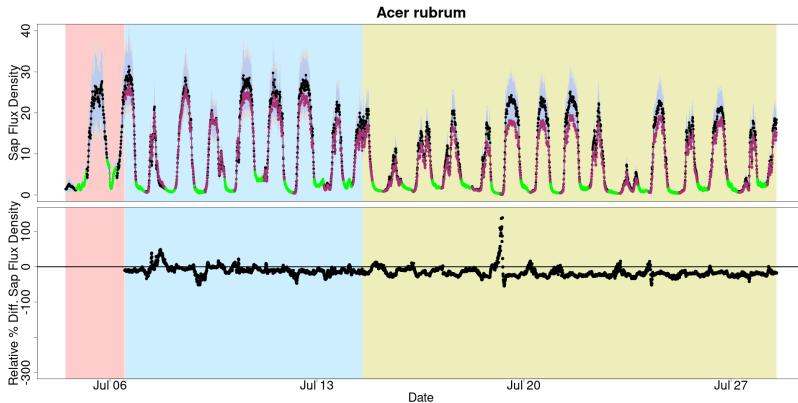


Australian Wildfires

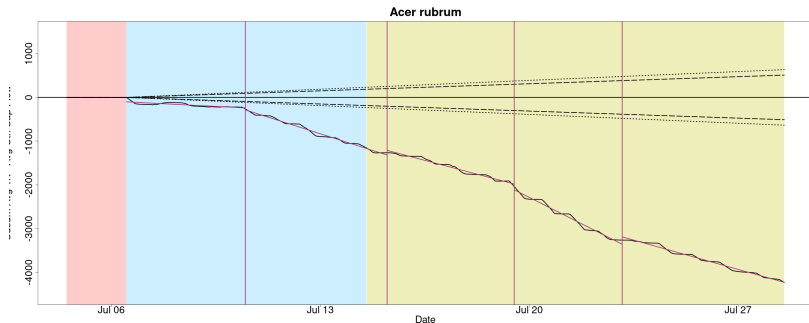


Left=score, Right=timing

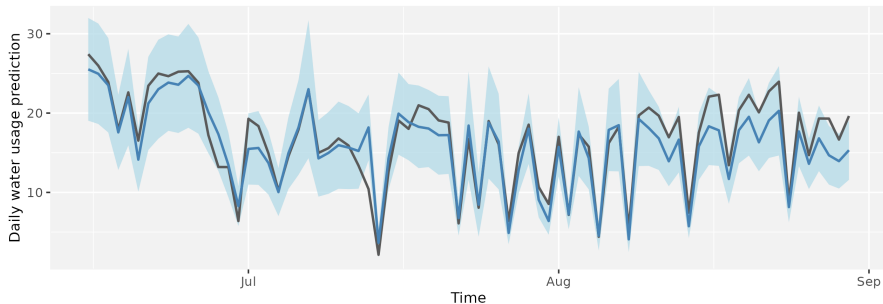
Tree Waterlogging



Tree Waterlogging

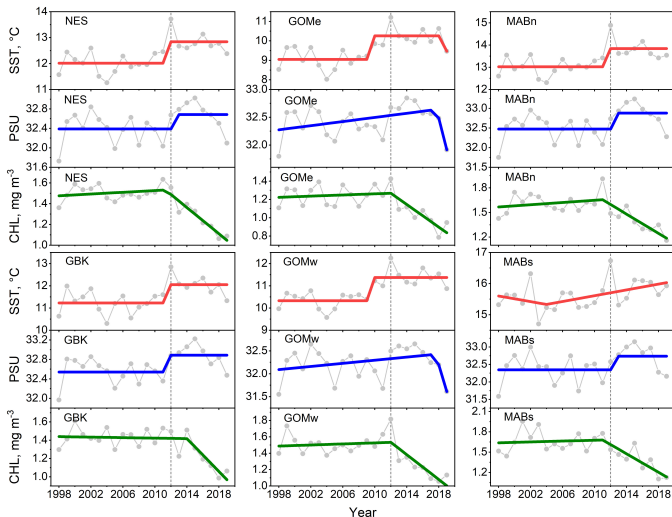


Predict Water Use



Predict whole season in advance, black=observed.

Based on SAP flux or dendromeda data.



Friedland et al. 2020 doi:10.1371/journal.pone.0028945

- Changepoints are an interesting area of research
- Changepoints occur in a variety of data
- The field has advanced considerably in recent years
- If you are interesting in applying it to some of your own data ...
- ... check out the changepoint R and ruptures Python packages, or reach out.